

Q1 Mutable vs immutable objects.

Ans ~~Mutable~~ vs Immutable objects are as

Mutable objects can be changed after creation (e.g., list, dict, set).

Immutable object cannot be changed once created (e.g., int, float, str, tuple)

Q2 5 common string operations

Ans len (string) - returns length of the string

string.lower() - converts to lowercase.

string.upper() - converts to uppercase.

string.replace (old, new) - replaces substring

string.split (delimiter) - splits string into a list.

Q3 Difference between list, tuple, set, dictionary

Difference Between list, tuple, set, dictionary

- any

Ans

Type	Ordered	Mutable
list	Yes	Yes
Tuple	Yes	No
Set	No	Yes
Dictionary	No	Yes

Q4

How to access dictionary elements.

Using Keys:- my_dict ['key']

Using get () method

my_dict.get('key')

Common list methods (append, insert, remove)

append(item) - adds item to end.

insert(index, item) - inserts item at index

remove(item) - removes first occurrence of item

Coding

Q1 Reverse a string

```
S = "hello"
```

```
reversed_s = S[::-1]
```

```
print(reversed_s)
```

Output:

'olleh'

Q2 Remove Duplicates from a List Using a Set

```
my_list = [1, 2, 2, 3, 4, 4, 5]
```

```
unique_list = list(set(my_list))
```

```
print(unique_list)
```

Output:

[1, 2, 3, 4, 5] (order may vary)

Q3 Dictionary of Students and Marks; Print Students with Marks > 75

```
Students = {
```

```
    "Alice": 82,
```

```
    "Bob": 67
```

```
    "Charlie": 90
```

"David": 74

```
{
  for name, marks in students.items():
    if marks > 75
      print(name)
```

Output
Alice
charlie